AD 2. AERODROMES

VAHS AD 2.1 AERODROME LOCATION INDICATOR AND NAME

VAHS - RAJKOT INTERNATIONAL AIRPORT, RAJKOT

VAHS AD 2.2 AERODROME GEOGRAPHICAL AND ADMINISTRATIVE DATA

1	Aerodrome reference point coordinates and its site	222318N 0710246E 269.41 DEG/ 475 M from physical extremity of RWY 23	
2	Direction and distance of aerodrome reference point from the center of the city or town which the aerodrome serves	30km North-east of Rajkot City	
3	Aerodrome elevation and reference temperature	650 FT / 41.7 DEG C	
4	Magnetic variation, date of information and annual change	0.25 DEG W (2010) /0.033 DEG E	
5	Name of aerodrome operator, address, telephone, telefax, e-mail address, AFS address, website (if available)	Airports Authority of India Airport Director Opposite NH 8A, Near Beti River Rajkot International Airport Rajkot (Gujarat) PIN Code: 360023,	
		Telephone:	(91) 281 2992328
		Fax:	
		AFS:	
		Email:	apdvahs@aai.aero
6	Types of traffic permitted (IFR/VFR)	IFR/VFR	
7	Remarks	i. Gravity Model: EGM-08	

VAHS AD 2.3 OPERATIONAL HOURS

1	Aerodrome Operator	MON-FRI: 0400-1300 UTC SAT/SUN/HOL: NIL
2	Custom and immigration	NIL
3	Health and sanitation	During AD Hours
4	AIS briefing office	As ATS
5	ATS reporting office (ARO)	As ATS
6	MET Briefing office	As ATS
7	Air Traffic Service	Refer NOTAM
8	Fuelling	As ATS
9	Handling	As ATS
10	Security	As ATS
11	De-icing	NIL
12	Remarks	ATS can be made available with 24-hour prior notice for other than notified periods.

VAHS AD 2.4 HANDLING SERVICES AND FACILITIES

1	Cargo-handling facilities	NIL
2	Fuel and Oil types	JET A1

3	Fuelling facilities and capacity	Hydrant fueling system: NIL Fuel service providers: IOCL and RBML Bowser fueling capacity: IOCL 2 bowsers Capacity - 9KL and 6KL RBML 4 bowsers Capacity: 11KL for one bowser and 16KL for remaining three.
4	De-icing facilities	NIL
5	Hangar space for visiting aircraft	NIL
6	Repair facilities for visiting aircraft	NIL
7	Remarks	Nil

VAHS AD 2.5 PASSENGER FACILITIES

1	Hotel(s) at or in the vicinity of aerodrome	In the city
2	Restaurant(s) at or in the vicinity of aerodrome	In the city
3	Transportation possibilities	Taxi, Public Transport
4	Medical Facilities	First Aid kit at Aerodrome, Hospitals in the city
5	Bank and post office at or in the vicinity of aerodrome	Banks: In the City Post office: In the City
6	Tourist office	In the City
7	Remarks	Nil

VAHS AD 2.6 RESCUE AND FIRE FIGHTING SERVICES

1	Aerodrome category for fire fighting	Within ATS HR: CAT-7
2	Rescue equipment	Available as per category
3	Capability for removal of disabled aircraft	1. The Critical Aircraft of Airport is A321. 2. Primary Responsibility for removal of disabled aircraft rests with the Airport Director Rajkot International Airport. 3. The airlines may contact 'Aerodrome Coordinator for Disabled Aircraft Removal Operations 'for coordinating with the agencies having Disabled Aircraft Recovery Kit. 4. Contact Details of Aerodrome Coordinator for Disabled Aircraft Removal Operations: Aerodrome Coordinator: Airport Director Email: apdvahs@aai.aero Contact No.: +91-9427207439
4	Remarks	Nil

VAHS AD 2.7 SEASONAL AVAILABILITY CLEARING

1	Type(s) of clearing equipment	NIL
2	Clearance priorities	NIL
3	Remarks	NIL

VAHS AD 2.8 APRONS, TAXIWAYS AND CHECK LOCATIONS/POSITIONS DATA

1	Designation, surface and strength of aprons	Refer Aircraft Parking/Docking Chart
2	Designation, width, surface and strength of taxiways	Refer Aircraft Parking/Docking Chart
3	Location and elevation of altimeter checkpoints	Location: On Apron Elevation: Refer Aircraft Parking/Docking Chart
4	Location of VOR checkpoints	On TWY A1 and TWY A2

5	Position of INS checkpoints	NIL
6	Remarks	**1. Runway Holding Position:** i. TWY A1: 222240.65319 N 0710213.29223 E; ELEV: 615 FT ii. TWY A2: 222235.70484 N 0710207.36078 E; ELEV: 609 FT iii. TWY A3: 222214.32177 N 0710141.75453 E; ELEV: 586 FT iv. TWY C: 222218.49000 N 0710156.17000 E; ELEV: 596 FT **2. Runway-Taxiway Intersection Point:** i. TWY A1: 222238.476 N 0710215.395 E; ELEV: 619 FT ii. TWY A2: 222233.524 N 0710209.465 E; ELEV: 613 FT iii. TWY A3: 222212.143 N 0710143.859 E; ELEV: 590 FT iv. TWY C: 222220.666 N 0710154.068 E; ELEV: 599 FT 3. Gravity Model: EGM-08

VAHS AD 2.9 SURFACE MOVEMENT GUIDANCE AND CONTROL SYSTEM AND MARKINGS

1	Use of aircraft stand identification signs, taxiway guidelines and visual docking/parking guidance system at aircraft stands	Taxiing guidance provided on R/T. AVDGS with aerobridge facility AVBL on bay no. 02, 03,04 and 05.
2	Runway and taxiway markings and lights	RWY Markings: Designation, THR, TDZ, Centerline, Aiming Point Marking. Lights: THR, Edge, End, Centreline TWY Marking: Centerline, Holding Positions, Edge Lights: Edge
3	Stop bars (if any)	Stop bar lights are provided at Taxi A1, A2, A3 and C.
4	Remarks	NIL

VAHS AD 2.10 AERODROME OBSTACLES

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM -08)	Marking/LGT	Remarks
In circling area and at AD	SIGN	222210.9N 0710144.5E	593 FT	NIL	Sign Board
In circling area and at AD	SIGN	222217.5N 0710152.2E	598 FT	NIL	Sign Board
In circling area and at AD	SIGN	222215.3N 0710145.7E	594 FT	NIL	Sign Board
In circling area and at AD	SIGN	222303.7N 0710247.8E	646 FT	NIL	Sign Board
In circling area and at AD	NAVAID	222306.8N 0710255.8E	658 FT	NIL	Glide Path LP DME (L) Top
In circling area and at AD	OTHER	222306.8N 0710255.7E	695 FT	NIL	Lightning Arrester on Glide Path Main Antenna
In circling area and at AD	NAVAID	222306.8N 0710255.7E	691 FT	NIL	Glide Path Main Antenna Top

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM -08)	Marking/LGT	Remarks
In circling area and at AD	OTHER	222308.8N 0710253.7E	647 FT	NIL	PAPI
In circling area and at AD	OTHER	222308.8N 0710258.1E	661 FT	NIL	Lightning Arrester on Glide Path Monitor Antenna
In circling area and at AD	NAVAID	222308.8N 0710258.1E	658 FT	NIL	Glide Path Monitor Antenna Top
In circling area and at AD	SIGN	222310.3N 0710255.7E	650 FT	NIL	Sign Board
In circling area and at AD	SIGN	222313.0N 0710254.8E	648 FT	NIL	Sign Board
In circling area and at AD	OTHER	222314.3N 0710252.2E	675 FT	NIL	Lightning Arrester on MET Antenna (MET Antenna Top = 204.9m)
In circling area and at AD	SIGN	222316.7N 0710303.4E	653 FT	NIL	Sign Board
In circling area and at AD	SIGN	222243.9N 0710224.1E	629 FT	NIL	Sign Board
In circling area and at AD	SIGN	222257.1N 0710239.9E	642 FT	NIL	Sign Board
In circling area and at AD	SIGN	222250.5N 0710232.0E	637 FT	NIL	Sign Board
In circling area and at AD	OTHER	222414.6N 0710138.3E	885 FT	NIL	Cellphone Mast (Light Top = 267.2 M.)
In circling area and at AD	OTHER	222251.8N 0710750.4E	1184 FT	NIL	Communication Tower (Hill Top = 333.0 M.)

VAHS AD 2.11 METEOROLOGICAL INFORMATION PROVIDED

1	Name of the associated meteorological office	Rajkot (Class 3)
2	Hours of service and, where applicable, the designation of the responsible meteorological office outside these hours	H24
3	Office responsible for preparation of TAFs and periods of validity and interval of issuance of the forecasts	Ahmedabad, 9 Hours
4	Availability of the trend forecast for the aerodrome and interval of issuance	NIL
5	Information on how briefing and/or consultation is provided	NIL
6	Types of flight documentation supplied and language(s) used in flight documentation	Tabular form English
7	Charts and other information displayed or available for briefing or consultation	NIL
8	Supplementary equipment available for providing information on meteorological conditions, e.g. weather radar and receiver for satellite images;	Telephone

9	The air traffic services unit(s) provided with meteorological information	Hirasar ATC and ACS
10	Additional information, e.g. concerning any limitation of service.	Documentation provided O/R. WDI available and lighted.

VAHS AD 2.12 RUNWAY PHYSICAL CHARACTERISTICS

Designations	TRUE Bearings	Dimensions of RWY (M)	Strength of pavement (PCN) and associated data) and surface of runway and associated stopways	Geographical coordinates for threshold and runway end
1	2	3	4	5
05	48.09 DEG	3040 x 45 M	99/R/B/W/T Concrete	THR: 222211.89N 0710143.56E RWY END: 222317.91N 0710302.64E
23	228.09 DEG	3040 x 45 M	99/R/B/W/T Concrete	THR: 222317.91N 0710302.64E RWY END: 222211.89N 0710143.56E

THR elevation and highest elevation of TDZ of precision APP RWY	Slope of runway and associated stopway	Dimensions of stopway (M)	Dimensions of clearway (M)	Dimensions of strips (M)
6	7	8	9	10
THR: 590.0FT TDZ: 610.0FT		NIL	NIL	3160 x 280 M
THR: 650.0FT TDZ: 650.0FT		NIL	NIL	3160 x 280 M

Dimensions of runway end safety areas	Location and description of arresting system (if any)	Existence of an obstacle-free zone	Remarks.
11	12	13	14
240M x 90M			1. Slope of runway and associated stopway: 0 to 800 M: + 0.699 % 900 to1800 M: +0.788 % 2060 to 3040 M: + 0.347 % 2. Gravity Model: EGM 08
240M x 90M			1. Slope of runway and associated stopway: 0 to 980 M: -0.347 % 1240 to 2140 M: -0.788 % 2240 to 3040 M: -0.699 % 2. Gravity Model: EGM 08

VAHS AD 2.13 DECLARED DISTANCES

RWY Designator	Take-off run available TORA (M)	Take-off distance available TODA (M)	Accelerate distance available ASDA (M)	Landing distance available LDA (M)	Remarks (including runway entry or start point where alternative reduced declared distances have been declared)
1	2	3	4	5	6
05	3040	3040	3040	3040	
23	3040	3040	3040	3040	

VAHS AD 2.14 APPROACH AND RUNWAY LIGHTING

Runway Designator	Type, length and intensity of approach lighting system	Runway threshold lights, colour and wing bars	Type of visual slope indicator system	Length of runway touchdown zone lights
1	2	3	4	5
05	SALS 420 M LIH	Green AVBL	PAPI LEFT/3.00 DEG MEHT (15.88M)	
23	CAT I 900 M LIH	Green AVBL	PAPI LEFT/3.00 DEG MEHT (16.91M)	

Length, spacing, colour and intensity of runway centre line lights	Length, spacing, colour and intensity of runway edge lights	Colour of runway end lights and wing bars	Length and colour of stopway lights	Remarks
6	7	8	9	10
3040 M 15 M LIH White	3040 M 60 M White LIH	Red		NIL
3040 M 15 M LIH White	3040 M 60 M White LIH	Red		NIL

VAHS AD 2.15 OTHER LIGHTING, SECONDARY POWER SUPPLY

1	Location, characteristics and hours of operation of aerodrome beacon/identification beacon (if any)	ABN	Top of ATC Tower 24 Flashes per minute Alternate White and Green
		IBN	Not Available
2	Location and lighting (if any) of anemometer/ landing direction indicator;	LDI	In front of Fire emergency road (LHS)
		Anemometer	Not Available
3	Taxiway edge and taxiway centre line lights;	Edge	A, A1, A2, A3, B1, B2 & C. Colour - Blue.
		Centre Line	NIL

4	Secondary power supply including switch-over time;	Secondary Power supply to all lighting at AD. Switch-over time: All RWY Lights: 01 SEC. : All TWY Lights: 01 SEC. : For other Lights:15 SEC.
5	Remarks	Apron Edge light available. Turn Pad Edge light – RWY 23 in blue colour available.

VAHS AD 2.16 HELICOPTER LANDING AREA

1	Geographical coordinates of the geometric centre of touchdown and lift-off (TLOF) or of each threshold of final approach and take-off (FATO) area	Not established
2	TLOF and/or FATO area elevation:	Not established
3	TLOF and FATO area dimensions to the nearest metre or foot, surface type, bearing strength and marking;	Not established
4	True bearings of FATO;	Not established
5	Declared distances available	Not established
6	Approach and FATO lighting;	Not established
7	Remarks	Not established

VAHS AD 2.17 AIR TRAFFIC SERVICE AIRSPACE

1	Airspace designation, geographical coordinates and lateral limits	CTR: Circular area centered on 222318N 0710246E within a 25NM radius.
2	Vertical limits	FL 65
3	Airspace classification	D
4	Call sign and language(s) of the air traffic services unit providing service;	Rajkot Tower, English
5	Transition altitude	4000 FT
6	Hours of applicability	HO
7	Remarks	Rajkot Approach Control Unit is combined with Rajkot Aerodrome Control Tower.

VAHS AD 2.18 AIR TRAFFIC SERVICES COMMUNICATION FACILITIES

Service Designation	Call sign	Channel(s)	SATVOICE Number(s), if available
1	**2**	**3**	**4**
APP	Rajkot Tower	120.025 MHZ	
APP	Rajkot Tower	120.275 MHZ	
TWR	Rajkot Tower	120.025 MHZ	
TWR	Rajkot Tower	120.275 MHZ	

Logon address, as appropriate	Hours of operation	Remarks
5	**6**	**7**
	HO	1. MAIN FREQ 2.Functions of Approach Control Service provided by Aerodrome Control Tower.

Logon address, as appropriate	Hours of operation	Remarks
	HO	1. BACKUP FREQ 2. Functions of Approach Control Service provided by Aerodrome Control Tower.
	HO	1. MAIN FREQ
	HO	1. BACKUP FREQ

VAHS AD 2.19 RADIO NAVIGATION AND LANDING AIDS

Type of aids, magnetic variation and type of supported operation for ILS/ MLS, basic GNSS, SBAS and GBAS, and for VOR/ILS/MLS station used for technical lineup of the aid	Identification	Frequency(ies), Channel number(s), Service provider, and reference path identifier(s) (RPI), as appropriate	Hours of operation, as appropriate;
1	2	3	4
LOC 23	IHSR	109.700 MHz	HO
GP 23		333.200 MHz	HO
DME ILS 23	IHSR	CH34X	HO
DVOR/DME	HSR	116.500 MHz CH112X	HO

Geographical coordinates of the position of the transmitting antenna	Elevation of transmitting antenna of DME/ elevation of GBAS reference point	Service volume radius from the GBAS reference point	Remarks
5	6	7	8
222204.9N 0710135.2E			
222306.8N 0710255.7E			1. Glide Path angle 3 Degree
222306.7N 0710255.8E	663 FT		1. Collocated with GP RWY23 2. Vertical Reference System: EGM-08
222208.0N 0710152.0E	625 FT		1. Vertical Reference System: EGM-08 2. DME colocated with DVOR

VAHS AD 2.20 LOCAL AERODROME REGULATIONS

• No Aircraft shall take lock turn/ pivotal turn on the Runway to avoid pavement damage.

VAHS AD 2.21 NOISE ABATEMENT PROCEDURES

NIL

VAHS AD 2.22 FLIGHT PROCEDURES

LOW VISIBILITY PROCEDURES
RAJKOT INTERNATIONAL AIRPORT, RAJKOT (VAHS)

1.INTRODUCTION:

The Low Visibility Procedure (LVP) incorporates safeguarding measures to mitigate runway incursions and defines operational restrictions to ensure safe aircraft operations taking into account the available Aerodrome facilities. This LVP is applicable only for take- off operations at Rajkot International Airport during conditions of Visibility/ RVR less than 800M/ 550M but not less than 350M. (LVP procedures shall not be applied at Rajkot airport if RVR is less than 350m). These procedures are required for compliance with DGCA Civil Aviation Requirements Series C Part I Section 8 Para 5.3.

2.DEFINITIONS & ABBREVIATIONS

2.1Instrument approach operations: An approach and landing using instruments for navigation guidance based on an instrument approach procedure. There are two methods for executing instrument approach operations:

a)A two-dimensional (2D) instrument approach operation, using lateral navigation guidance only; and

b)A three-dimensional (3D) instrument approach operation, using both lateral and vertical navigation guidance

Note – Lateral and vertical navigation guidance refers to the guidance provided either by:

i)a ground-based radio navigation aid; or

ii)Computer-generated navigation data from ground-based, space-based, self-contained navigation aids or a combination of these.

Instrument approach operations shall be classified based on the designed lowest operating minima below which an approach operation shall only be continued with the required visual reference as follows:

Type A: a minimum descent height or decision height at or above 75m (250ft); and

Type B: a decision height below 75 m (250 ft.). Type B instrument approach operations are categorized as:

Category I (CAT I): a decision height not lower than 60 m (200 ft.) and with either a visibility not less than 800 m or a runway visual range not less than 550m;

Category II (CAT II): a decision height lower than 60 m (200 ft.) but not lower than 30 m (100 ft.) and a runway visual range not less than 300 m;

Category III (CAT III): a decision height lower than 30 m (100 ft.) or no decision height and a runway visual range less than 300 m or no runway visual range limitations.

2.2Decision Altitude/Height: A specified altitude or height in a 3D instrument approach operation at which a missed approach must be initiated if the required visual reference to continue the approach has not been established.

Note. - Decision altitude (DA) is referenced to mean sea level and decision height (DH) is referenced to the threshold elevation.

2.3ILS Critical Area: An area of defined dimensions about the localizer and glide path antennas where aircraft and vehicles are excluded during all ILS operations. The critical area is protected because the presence of vehicles or aircraft inside its boundaries will cause unacceptable disturbance to the ILS signal-in-space.

2.4ILS Sensitive Area: An area extending beyond the ILS critical area where the parking and/or movement of vehicles, including aircraft, are controlled to prevent the possibility of unacceptable interference with the ILS signal during ILS operations. The sensitive area is protected to provide protection against interferences caused by large moving objects outside the critical area but still normally within the airfield boundary.

2.5Low Visibility Procedures: Specific procedures applied at an aerodrome for the purpose of ensuring safe operations during Categories II/ III approaches and/or low visibility take-offs.

2.6Low Visibility Take-Off (LVTO): A term used in relation to flight operations referring to a take-off on a Runway where the RVR is less than 400m.

2.7Manoeuvring Area: That part of an aerodrome to be used for the take-off, landing and taxiing of aircraft, excluding aprons.

2.8Movement Area: That part of an aerodrome to be used for the take-off, landing and taxiing of aircraft, consisting of the manoeuvring area and the apron(s).

2.9Visibility: Visibility for aeronautical purposes is the greater of:

a) The greatest distance at which a black object of suitable dimensions, situated near the ground, can be seen and recognized when observed against a bright background;

b) The greatest distance at which lights in the vicinity of 1 000 candelas can be seen and identified against an unlit background

Note 1. — the two distances have different values in air of a given extinction coefficient, and the latter b) varies with the background illumination. The former a) is represented by the meteorological optical range (MOR).

Note. 2. — the definition applies to the observations of visibility in local routine and special reports, to the observations of prevailing and minimum visibility reported in METAR and SPECI and to the observations of ground visibility.

2.10Runway Visual Range: The range over which the pilot of an aircraft on the centreline of a runway can see the runway surface markings or the lights delineating the runway or identifying its centre line

2.11Aerodrome Operating Minima: The limits of usability of an aerodrome for:

a) take off, expressed in terms of runway visual range and / or visibility and, if necessary, cloud conditions

b) landing in 2D instrument approach operations, expressed in terms of visibility and/or runway visual range; minimum descent altitude/height (MDA/H) and, if necessary, cloud conditions; and

c) Landing in 3D Instrument Approach operations, expressed in terms of visibility and/or runway visual range and decision altitude/height (DA/H) appropriate to the type and/or category of the operation

3.ABBREVIATIONS

ADC	Aerodrome control
ADP	Airside driving permit
ARFFS	Airport rescue and firefighting services.
ATC	Air Traffic Control
ATIS	Automatic terminal information services
ATM	Air Traffic Management
AVP	Airside vehicle Permit
CAR	Civil Aviation Requirements
CAT	Category
CFT	Crash Fire Tender
CISF	Central Industrial Security Force
DG	Diesel Generator
DGCA	Directorate general civil Aviation
DH	Decision Height
ICAO	International Civil Aviation Organisation
ILS	Instrument Landing system
GSECL	Gujrat state electricity corporation limited
LVP	Low visibility procedures
LVTO	Low visibility take-off
MDA	Minimum descent altitude
MEL	Minimum Equipment Limit
MET	Meteorology
PAPI	Precision Approach Path Indicator
PDC	Pre-determined position
PIC	Pilot in command
RT	Radio Telephony
RVR	Runway visual range
SMC	Surface Movement control
SOP	Standard operating procedures
SP	Safeguarding Procedure
TDZ	Touchdown zone
TWR SUP	Tower Supervisor
TWY	Taxiway

4.MINIMUM FACILITIES REQUIRED TO SUPPORT LVP

The following Aeronautical Ground lights/equipment shall be serviceable to the
required standard to support Low Visibility Procedures. In the absence of any of the following facilities, the LVP operation shall be suspended.

a) Runway Edge Lights- Switch overtime: 1 sec
b) Runway End Lights- Switch over time: 1sec
c) Runway Centre line Lights- Switch over time: 1sec
d) TWY edge lights, Runway guard lights - Switch over time: 15sec,
e) Secondary/stand by power supply to meet above specified switch over time requirements.
f) DCWIS system is installed in MET Park (RWY 23 side) and FSM (Forward Scattered Meter) equipment is used for providing instantaneous RVR for the RWY 23.
Note: Mid RVR and Roll- out RVR are not required for LVP as no operation is permitted below 350m RVR at Rajkot International Airport.
g) NAV AIDS

Equipment Name	Facility	Frequency and ID	Location co-ordinates / location of the facility
LOC	NAV AID	109.7 MHz IHSR	Near RWY 05 222204.941N 0710135.237E
GP	NAV AID	333.2MHz IHSR	Near RWY 23 222306.837N 0710255.707E
LP DME	NAV AID	995/1058 MHz IHSR	222306.775N 0710255.791E
DVOR	DVOR TERMINAL	116.500 MHz HSR	222208.037N 0710152.000E
HP DME	NAV AID	1136/1199 MHz HSR	222207.947N 0710152.349E

Note: ILS critical and sensitive area for Rajkot International Airport is provided in Annexure B.

5.SERVICEABILITY REQUIREMENTS OF AERONAUTICAL GROUND LIGHTS BEFORE IMPLEMENTATION OF LVP (REF. DGCA CAR 4B-1 /ANNEX 14, CLAUSE 10.4.11)

Aeronautical Ground Lighting system	Un-serviceability	Restriction
RWY Edge/Centre line/Taxiway lights	More than 5%of all lights are unserviceable	LVP operations will be Suspended
	Any two consecutive lights or more unserviceable	
Runway Threshold lights/End lights	More than 25% of all lights are unserviceable.	LVP operations will be Suspended.
	Any two consecutive lights or more are unserviceable	
DG set for power supply to AGL in case of unserviceability of the main power supply	Any DG set supplying power to the AGL/ equipment is unserviceable	LVP operations will be suspended
RVR System	Automated RVR system unserviceable	LVTO will be continued if manual RVR can be provided

Details of AGL installation at Rajkot International Airport is provided in Annexure A.

5.1NOTE:

a) If the intensity of any lights has reduced more than 50 % of its minimum required level, the light shall be considered unserviceable.

b) There is NO system at Rajkot that automatically relays to maintenance crew or air traffic control unit about any unserviceability of aeronautical ground lighting system as prescribed in para 8.3.4 and 8.3.5 of CAR section 4 series B part 1.

c) In accordance with para 10.5.11 of CAR section 4 series B part 1 manual runway inspection is carried out with electrical engineering before implementation of LVP to check minimum serviceability of aeronautical ground lighting system.

6.

SAFEGUARDING PROCEDURES

Safeguarding Procedures (SP) are instructions for airport departments, Airline operators, ground handling agents and other agencies working in the airside to prepare ground services and facilities for low visibility operations in order that LVP are implemented after all Safeguarding Procedures (SP) are completed.

Duty Officer Tower/Tower Supervisor will initiate and co

-coordinate with all the concerned agencies for completion of safeguarding procedures before implementation of Low Visibility Procedures.

6.1Safeguarding Procedures shall be initiated when:

The Visibility/RVR is less than 1200m and the forecast is to deteriorate to 800m or less.

6.2 Safeguarding procedures include:

a) Runway inspection shall be carried out by the Ops Incharge on duty along with electrical engineering officials before implementation of LVP.

b) Positioning of one CFT each at the two predetermined positions (ATC approach road outside the Runway strip and near Fire Station).

c) Stopping of all maintenance works on the manoeuvring area as well as removal of all men and mobile equipment from the said area including RWY strip.

d) Safeguarding the Runway from incursion via various access road to the runway. This may be achieved by removing all non-essential vehicles from the operational area under supervision of Ops Incharge.

e) Ensuring uninterrupted power supply for essential facilities using DG and taking main power supply as standby.

f) Restrict all vehicular movement near ILS critical and sensitive area.

7.LOW VISIBILITY PROCEDURES

7.1Implementation of Low Visibility Procedures:

Tower Duty officer and Tower A shall monitor the deterioration of visibility to 800 Meters or below Tower Supervisor/ Tower A shall coordinate with all the agencies to confirm whether the Safeguarding procedures have been completed. When Visibility/RVR falls below 800M/550M and safeguarding procedures are completed, Tower supervisor will implement Low Visibility Procedures. Duty Officer Tower shall inform ARFF and Ops Incharge for the implementation of low visibility procedures.

7.2Restriction on vehicular movements:

The vehicles on the manoeuvring area shall be restricted to bare minimum during the

period of LVP. All vehicles previously operating on the manoeuvring area shall be vacated and parked away from the RWY strip preferably on the perimeter road in the valley area. Ops Incharge shall monitor and keep the record of all such vehicles

8.ACTIONS BY VARIOUS AGENCIES DURING LVP

8.1 CONTROL TOWER

a) Duty Officer, TWR should ensure the appropriate Aeronautical ground lights must have been inspected during the hour preceding implementation of LVP, and thereafter once every two-hour period to ensure the serviceability of all Aeronautical Ground Lights. These inspections of Aerodrome Ground Lights should be accorded priority and, if necessary, aircraft operations may be delayed. Necessary entries in respect of any unserviceability of AGLs shall be made in the logbook (TWR D) kept in Tower.

b) ATC Tower shall inform Ops Incharge, Fire Station, and AGL about the implementation of low visibility operations.

c) Duty Officer Tower shall ensure the MET official is present in control Tower for real time update of Visibility/RVR.

d) Tower shall not permit any ground run of aircraft on the manoeuvring area

e) Number of Aircraft movement during LVP shall be restricted to one aircraft only.

f) Restrictions imposed by 9-member committee constituted to oversee the recommendations of AAIB regarding aircraft movement on aircraft stand taxi lane during landing and take- off shall be strictly followed. (No aircraft shall be on the aircraft stand taxi lane at apron during take-off).

g) The following may be included in ATIS broad cast (presently ATIS not available at Rajkot International Airport) "LOW VISIBILITY PROCEDURE IN FORCE".
h) TWR shall permit departures only from the beginning of the Runway in use.
i) If the reported RVR id less than 400m, TWR shall obtain the approved LVTO minima from the pilot.
•If RVR is obtained manually, ATC shall inform the Pilot that the observation is manual. PIC may make his own assessment of the RVR before departure to ensure that it is above 350M as update of Manual RVR may take time.
•Duty Officer Tower shall stop all departures whenever RVR falls below 350M.
•A take-off alternate aerodrome shall be asked by tower controller if either the meteorological conditions at Rajkot are below the operator's established aerodrome landing minima for that operation or if it would not be possible to return to the aerodrome of departure for other reasons.
8.2OPS INCHARGE
a) Inform all users of apron (airlines/ground handling agencies/stakeholders), Engg-Civil, Engg-Electrical, ARFF, CISF Control room, Equipment room and MT section regarding commencement of implementation of SGP.
b) Restrict the movement of vehicular traffic on movement area by advising CISF Control Room.
c) OPS Incharge shall coordinate with airline operators and ground handling agencies and take necessary action to safeguard the Aircraft stand taxi lane on the apron during the implementation of LVP.
d) No men / vehicles / equipment are allowed to cross the apron safety lane.
e) The procedure of SP/LVP shall be included in the ADP/AVP training
f) Inform Duty Officer Engg. (Elect) to check and confirm serviceability/ availability of generator for secondary power supply.
g) Inform Civil/ Electrical Maintenance Division to suspend all works in progress in operational area and removal of men, machinery and material from the movement area.
h) Inform all users of the Apron (airlines/other agencies) regarding enforcement of Low Visibility Procedure and confirm its compliance to ATC Tower.
i) As and when ATC notify for withdrawal of Safeguarding Procedure, all concerned shall be intimated.
j) Ensure availability of 'Follow Me' service to aircraft as and when requested by ATC.
k) No vehicle on Apron shall enter the manoeuvring area without permission from ATC tower.
l) No vehicle shall be permitted except Apron control vehicle fitted with Tower VHF R/T for RWY/electrical inspection. In case of any emergency requirement, Apron control jeep shall provide escort to other needy vehicles.
m) All Non-Sked aircraft shall be provided with Follow-Me service mandatorily and scheduled aircraft as and when requested.
n) Ensure that bird scares are properly distributed and briefed accordingly.
o) Ops Incharge shall inform ATC tower on completion of all safeguarding procedures for the implementation of LVP.
p) Ensure to restrict vehicular movement in ILS critical and sensitive area in coordination with CISF.
8.3ELECTRICAL ENGINEERING
a) Electrical In-charge shall ensure the availability/serviceability of Secondary Power Supply with change over time.
b) Inspection of AGL in co-ordination with Ops Incharge. Following visual aids associated with RWY 05/23 shall be serviceable as given in Para 5 & 5.1 above and can be operated at full intensity.
•Runway Edge lights
•Runway threshold and Runway end lights
•Taxiway edge lights
•Runway holding position lights & signs
•Runway guard lights
•Runway Center line lights
•Obstacle lights
Note: No adjustment of light intensities shall be made without permission from Control Tower.
c) Inform the serviceability of above visual lighting aids to ATC.
d) Ensure that Switch Room and New Powerhouse is manned. If one second switch over time cannot be achieved due to any reasons, Switch on generator set so as to run parallel with the primary source (GSECL) to meet requirement of switch over time as specified in para 5. The generator may be switched off when advised by apron control/ ATC.
e) Inform the un-serviceability or any change in the status of any facility/systems to ATC and Ops Incharge immediately.
8.4 AIRPORT RESCUE & FIRE FIGHTING SERVICE (ARFFS)
a) On receipt of notification of implementation of Safeguarding procedures/LVP, Airport Rescue and Fire Fighting Service (ARFFS) will remain in readiness at Fire Station and PDPs.

b) In the event of an incident when LVP is in force, positive clearance needs to be obtained by ARFFS from ATC prior for entering manoeuvring area. ATC and Ops Incharge should provide the maximum assistance in directing ARFFS to required location.

8.5CIVIL ENGINEERING

On receipt of notification from ATC/ Ops Incharge that safeguarding procedure is being implemented, Engg-Civil shall confirm the following to Ops Incharge.

a) Stop the work in progress in the operational area.

b) Remove men, machinery and material from the work site and leave the operational area to the valley side.

c) Resuming of work shall only be initiated after obtaining positive clearance from ATC.

8.6CISF

On receipt of notification from ATC/Apron Control/AOCC that safeguarding procedure is being implemented, CISF Control Room shall initiate the following actions:

a) Inform all access gates leading to operational area and CISF Posts in operational area to STOP entry of non-essential vehicles.

b) No vehicle from the CISF barrack shall enter the operational area without prior permission from ATC /coordination with Ops Incharge.

c) Ensure to restrict vehicular movement in ILS critical and sensitive area in coordination with ops InCharge

8.7AIRLINE OPERATORS/GROUND HANDLING AGENCIES/OTHER AGENCIES

a) All the agencies shall ensure that staff and drivers/operators are suitably trained and aware about the importance and prevailing conditions of Low Visibility operations, aerodrome lay-out, markings and signs in the operational area.

b) All agencies operating in the operational area shall ensure that only those vehicles that are absolutely essential for aircraft operations operate in the operational area during periods of low visibility. The drivers/operators of these vehicles should keep a look out for taxiing aircraft and other vehicles to prevent accidents.

c) All the vehicles must have their obstruction lights "ON" during operation of low visibility procedures. No vehicle without serviceable obstruction light shall be permitted in the apron/ manoeuvring area during LVP implementation.

d) All instructions/sign boards provided for vehicular movement area/service roads, must be followed while operating in the operational area. All vehicles shall operate at 10kmph or below during LVP on the apron.

e) No vehicle/equipment/ men shall cross the apron safety line demarcating the boundary of the aircraft stand taxi lane on the apron during LVP period.

f) Follow all instructions/sign boards provided for vehicular movement area/service roads.

g) No vehicle/equipment/personnel shall enter in and around the vicinity of the runways or taxiways without prior permission of ATC.

h) Airline operators shall ensure that the aircraft is suitably equipped and the pilots have necessary training and authorization to operate during LVP.

i) Every year before commencement of monsoon/winter season, a refresher/ sensitization program for personnel responsible for airside operations shall be conducted to create awareness and to ensure cooperation for safe airport operations during periods of low visibility.

8.8PILOTS

a) Pilot in command shall not initiate Take off below 800m visibility / 550m RVR unless the LVP are enforced.

b) The pilot shall ensure that the aircraft is suitably equipped for the operation during LVP meeting the minimum requirements including MEL as specified in the DGCA CAR.

c) If any time the reported visibility is below 800m, then RVR criteria shall be applicable for aerodrome operating minima. The RVR may be derived from an automated RVR system/manual observation of RVR.

d) If RVR is obtained manually, ATC shall inform the Pilot that the observation is manual. PIC may make his own assessment of the RVR before departure to ensure that it is above 350M as update of Manual RVR may take time.

e) The airline operator/ pilot in command shall ensure that the flying crew shall be meeting the training requirement specified by the regulator for the operation during LVP. The training criteria for the pilot has been specified in DGCA CAR section 8C part 1 - All weather Since Rajkot airport allows only departures during LVP, no cloud base minima is not considered for the implementation of LVP. Airline/ pilot in command has to follow their own minima based on cloud base if applicable.

f) For all departures during LVP, the pilot shall ensure that suitable alternate aerodrome/aerodromes are filed in their operational flight plan (Refer para 14.2 DGCA CAR- manual on of weather operations). Take off alternate shall be selected and specified in the OFP as per the criteria specified in para 14.2 (a) of DGCA CAR 8C part-1 (all weather operations).

g) Pilot in command shall ensure that they have the necessary approval for LVTO minima form DGCA/ regulatory authorities if the take-off is conducted below RVR 400M.

9.CONTINGENCY PROCEDURES

Degradation of RVR System

a) If automated RVR system is unserviceable, manual observation of RVR may be taken by MET department (IMD) if it is feasible, ATC may allow this and Ops Incharge shall make necessary arrangements for manual observation of RVR. If the manual RVR reported is 350m, the ATC may permit the departures.

b) The ATC while giving Manual RVR shall inform the pilot that the observation is manual. The time of observation shall also to be informed to the pilots.

Note: The requirement of Mid RVR / Roll out RVR is not required as the Take-off is not allowed below 350m RVR at Rajkot.

10.SUSPENSION/ TERMINATION OF LOW VISIBILITY PROCEDURES

a) When Visibility/RVR improves to 800M/550 M and trend is for further improvement; Tower Supervisor/Duty Officer Tower would terminate operations of LVP. He must obtain advice from Duty Officer, Meteorological Office regarding improvement in weather conditions before the termination of LVP. However, the runway safe guarding procedures will be continued till the visibility improve to 1200m or more

b) When the RVR falls below 350 meters the LVP will be suspended by Tower Supervisor/Duty Officer, Tower and no aircraft departures are allowed. However, runway safeguarding procedure will continue to be in force if the trend forecast indicates that the RVR is likely to be improved to 350 meters or above within a reasonable time.

c) The Aircraft Departures during LVP will be suspended by ATC in the following circumstances also.

•If the automated RVR system becomes unserviceable.

•If the facilities essential for the implementation of LVP becomes unserviceable (Ref. Para5& 5.1)

•If the runway lighting system failed to meet the minimum switch over time as specified in this SOP.

•Failure/ non-availability of secondary power supply.

•Failure of Runway lights monitoring system provided at control Tower

•Failure of two-way radio communication system of vehicles/ CFTs Deployed in operational area during LVP.

•If unable to safe guard the runway and Manoeuvring area during degrade mode of operation of Air Traffic services.

•On cancellation / Suspension of LVP, Duty officer Tower shall inform Ops Incharge and Fire station.

•Ops Incharge will inform all concerned agencies (Engg-Elect, Engg-Civil, Equipment room, MT section CISF and other stake holders. regarding the suspension/ termination of LVP operations.

•On cancelling/suspending of LVP, following message shall be included in two subsequent ATS broadcasts “LOW VISIBILITY PROCEDURES CANCELLED/ SUSPENDED “.

•If Safeguarding Procedures are implemented and LVP are not subsequently implemented and the visibility/RVR improves to 1200 m and above, then the Tower Supervisor/Duty Officer Tower may cancel Safe guarding Procedures in coordination with Ops Incharge

Annexure-A

Details of AGL installation at Rajkot International Airport

Sr. No.	Facility Description	Circuit Name	Fixture Type	Number of Fixture
1	Approach-23 CAT -I (Two Interleaving Circuits)	Approach 23, CAT - I (1A) (Interleaved Circuit)	Inset Elevated	1 61
2		Approach 23 CAT - I (1B) (Interleaved Circuit)	Inset Elevated	1 57
3	Approach-05 (SAPL) (Two Interleaving Circuits)	Simple Approach 05 (1A) (Interleaved Circuit)	Inset Elevated	1 18
4		Simple Approach 05 (1B) (Interleaved Circuit)	Inset Elevated	1 16
5	Runway Threshold & Threshold Wingbar-23 (Working	23 Threshold + Wing Bar (1) (Working)	Inset Elevated	16 10
6	Runway Threshold & Threshold Wingbar-23 (Standby)	23 Threshold + Wing Bar (2) (Standby)	Inset Elevated	16 10

7	Runway Threshold & Threshold Wingbar-05 (Working)	05 Threshold + Wing Bar (1) (Working)	Inset Elevated	6 10
8	Runway Threshold & Threshold Wingbar-05 (Standby)	05 Threshold + Wing Bar (2) (Standby)	Inset Elevated	6 10
9	Runway Centreline Light Circuit (Active, Interleaving)	Runway Centre line Light (1A) Interleaved Circuit	Inset	101
10		Runway Centre line Light (1B) Interleaved Circuit	Inset	101
11	Runway Edge Light + End + New turn pad (Working)	Runway Edge Light Circuit-1	Inset Elevate	10 96
		Runway End Light	Elevated	12
		Turn pad edge light (A)	Elevated	4
		Runway Designation Sign Board	LED	2
		Turning Pad Edge	Elevated	11
12	Runway Edge Light + End + new turn pad (Standby)	Runway Edge Light Circuit-2	Inset Elevated	13 95
		Runway End Light	Elevated	12
		Turn pad edge light (A)	Elevated	4
		Runway Designation Sign Board	LED	2
		Turning Pad Edge	Elevated	10
13	PAPI-23 (Two Interleaving Circuits)	23 PAPI (1A) (Interleaved Circuit)	Elevated	6
14		23 PAPI (1B) (Interleaved Circuit)	Elevated	6
15	PAPI-05 (Two Interleaving Circuits)	05 PAPI (1A) (Interleaved Circuit)	Elevated	6
16		05 PAPI (1B) (Interleaved Circuit)	Elevated	6
17	Turning Pad Centreline - 1 (Working)	Turning Pad Centreline (1) (Working)	Inset	47
18	Turning Pad Centreline - 2 (Standby)	Turning Pad Centreline (2) (Standby)	Inset	47
19	STOP BAR Taxi-A1 (Two Interleaving Circuits)	Stop Bar A1 (1A) Interleaved Circuit	Inset	6
20		Stop Bar A1 (1B) Interleaved Circuit	Inset	5
21	STOP BAR Taxi-A2 (Two Interleaving Circuits	Stop Bar A2 (1A) Interleaved Circuit	Inset	6
22		Stop Bar A2 (1B) Interleaved Circuit	Inset	5
23	STOP BAR Taxi-A3 (Two Interleaving Circuits)	Stop Bar A3 (1A) Interleaved Circuit	Inset	6
24		Stop Bar A3 (1B) Interleaved Circuit	Inset	5
25	STOP BAR Taxi C (Two Interleaving Circuits)	Stop Bar C (1A) Interleaved Circuit	Inset	5
26		Stop Bar C (1B) Interleaved Circuit	Inset	4
27	Runway Guard Light	Runway Guard Light (Working)	Elevated	8
28	Intermediate Holding Position (Taxiway A, B1, B2)	IHP Light (PTT) (Working	Inset	15

29	Taxiway-A1, A2, A3, A, B1 & B2 (Active, Interleaving)	Taxiway Edge Light (A/A1/A2/A3/ B1/B2) + Guidance Sign (1A) Interleaved Circuit	Elevated LED Signboard	110 41
30		Taxiway Edge Light (A/A1/A2/A3/ B1/B2) + Guidance Sign (1B) Interleaved Circuit	Elevated LED Signboard	113 41
31	Taxiway-C (Active, Interleaving	Taxiway Edge Light (C) + Guidance Sign (1A) Interleaved Circuit	Elevated LED Signboard	24 6
32		Taxiway Edge Light (C) + Guidance Sign (1B) Interleaved Circuit	Elevated LED Signboard	23 6
33	DTGM	Distance To Go Markers (Double Sided) 1-9	LED	9
	DTGM	Distance To Go Markers (Double Sided) 0/10	LED	2

VAHS AD 2.23 ADDITIONAL INFORMATION

NIL

VAHS AD 2.24 CHARTS RELATED TO AN AERODROME

1. Aerodrome Chart
2. Aircraft Parking/Docking Chart (11 JUN 2026)
3. Aerodrome Obstacle Chart Type A (Operating Limitations) RWY 05
4. Aerodrome Obstacle Chart Type A (Operating Limitations) RWY 23
5. Aerodrome Obstacle Chart Type B
6. Instrument Approach Chart - ILS or LOC Z Procedure RWY 23 (CAT A/B/C) (13 June 2024)
7. Instrument Approach Chart - ILS or LOC Y Procedure RWY 23 (CAT A/B/C) (13 June 2024)
8. Instrument Approach Chart - RNP Procedure RWY 23 (CAT A/B/C) (08 August 2024)
9. Tabular Description - RNP Procedure RWY 23 (CAT A/B/C) (08 August 2024)
10. Instrument Approach Chart - RNP Y Procedure RWY 05 (17 April 2025)
11. Tabular Description - RNP Y Procedure RWY 05 (17 April 2025)

VAHS AD 2.25 VISUAL SEGMENT SURFACE (VSS) PENETRATION

Procedure	Procedure Minima	VSS Penetration
1	**2**	**3**